DNA extraction from wood samples using CTAB buffers

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Sources:

- dx.doi.org/10.17504/protocols.io.bg6wjzfe
- With modifications from dx.doi.org/10.17504/protocols.io.bwr8pd9w and myself (who?)
- https://doi.org/10.1002/aps3.70028 - possible optimizations
- https://doi.org/10.1515/biol-2022-1006 - sodium acetate from here

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B-mercaptoethanol

Suggestions:

- Get second pair of gloves while working under the hood

Warnings:

- Always work under the hood
- Do not touch without gloves
- Do not inhale the fumes
- Clean hands with soap after usage

CTAB

Final volume: 500.0 ml

Reagents list:

- CTAB powder 10.0 g
- NaCl 40.908 g
- Tris-HCl 7.88 g
- EDTA 3.722 g

Preparation steps:

1. Add a magnetic stir bar to the autoclaved glass bottle.
2. Place the bottle on the stirrer/heating plate. Do not exceed 60 °C.
3. Add each component one by one, allowing each to dissolve completely before adding the next.
4. Bring the final volume to 500 ml with distilled water.
5. Continue stirring/heating until the solution is clear.

Suggestions:

- Dissolution may take up to a few days. Leave it on the heat, do not close the lid tight while mixing.

Proteinase K

Final volume: 1.0 ml

Reagents list:

- Proteinase K powder 20.0 mg
- Double-distilled autoclaved molecular-grade water 1.0 ml

Preparation steps:

1. Use an autoclaved and clean 1.5 ml tube.
2. Place the empty tube on the ultra-precision balance and tare the balance.
3. Add Proteinase K powder to the tube until approximately 20 mg (0.02 g) is reached.
4. Add 1 ml of double-distilled autoclaved molecular-grade water.
5. Mix until the Proteinase K is fully dissolved.

Suggestions:

- Add the Proteinase K powder gradually and reweigh the tube until approximately 20 mg is reached.

Phenol Chloroform Isomail

Warnings:

- Always work under the hood
- Do not touch without gloves
- Do not inhale the fumes
- Clean hands with soap after usage

Isopropanol

Sodium Acetate

Ethanol 70%

Ethanol 90%

Wood Homogenization

Suggestions:

- The typical workflow should be: enter the room, place/collect the sample, and leave ~30 sec

Warnings:

- Do not remain in the room during, after, or between homogenization runs.
- Chronic exposure to fine wood dust can seriously damage your lungs.

1. Setup homogenization mode: (5 minutes homogenization, 1 min rest) <- 4 cycles

2. Prepare/Check if container, spoon, beads are ready for homogenization.

Suggestions:

- Clean with soap and water if not.

3. Take the samples from the -80C freezer (beware the ultra cold temperatures).

4. Fill ~0.5 of container volume with wood material and beads.

Warnings:

- Make sure that the gasket between the lid and the container is in place

5. Then return the sample tube to -80C freezer

> **Suggestions:**
> - You can do it on the way to the lab with homogenizer.

6. Place the container while holding the lid in place, then tighten the clamp as much as possible

7. Close the homogenizer lid and press 'Start.'

8. After homogenization, take the container and move it under the hood with the airflow turned on

9. Transfer the contents into a clean 50 mL tube

10. Put the tube into the fridge and let it cool. After that, put under -80C

11. Clean the hood to remove any remaining wood particles, then increase the hood airflow. Reduce it again at step 4.

12. Repeat from step 2

Tissue lysis

> **Warnings:**
> - Work only under the hood on this stage
> - Make sure you have enought reagents and tubes
> - Avoid opening several tubes at the same time. Minimize cross-contamination.

1. Turn on the incubator, set temeprature to 65C

2. Prepare working solution

Suggestions:

- Take new 50ml tube
- Add 1200 ul per sample + 10% of CTAB buffer
- Add 20 ul per sample + 10% of BME
- Add 20 ul per sample + 10% of proteinase K

3. Add 1240ul of working solution to each sample

Suggestions:

- Add 620ul twice
- Use reverse pipeting technique to minimize 1000ul tip usage and increase precision

4. Close the tube, mix well by inverting the tube repeatedly

Warnings:

- Do not use the vortex since that can break the DNA fragments!

5. Take the tubes to the incubator and place them in the rotary plate with the rubber bands. Incubate the samples for 1h at 65C while rotating.

Suggestions:

- Label new 1.5ml tubes while waiting

Warnings:

- Make sure all tubes are VERY WELL CLOSED before placing them on the rotating plate and before starting rotation.
- Every couple of minutes, make sure the samples are mixing well (invert a few times if not).

6. Centrifuge all samples 10,000g for 15 minutes at room temperature, to precipitate the leftover wood and cell walls

7. Take the samples to the hood, set a 1000ul pipette to 640ul (more if you added more CTAB and other reagents), slowly transfer the supernatant to a clean, new 1.5ml tube already labelled

Suggestions:

- If you have more than 600ul of supernatant, you can split the volume into equal parts and separate tubes (e.g. you have 700ul, you can transfer 350ul to each tube for a total of two tubes - these will be replicates)

Warnings:

- Transfer as much liquid as possible while avoiding debris

8. Add 1 volume (if you have 640ul supernatant, add 640ul Ph:Ch:Is) of Phenol:Chloroform:Isoamyl alcohol, close the tube, and immediately mix by inverting the tube until everything is more or less homogenized

9. Centrifuge at 12,000g for a minimum of 30 minutes at room temperature. You should be able to see two liquid phases completely separated.

Suggestions:

- Label new 1.5ml tubes while waiting
- If the upper phase is cloudy, centrifuge for a bit longer.

10. Take the samples to the hood very carefully, avoid re-mixing the phases. Don’t shake the tubes or move them harshly.

11. Transfer the upper aqueous phase to a new 1.5ml tube labelled properly

Suggestions:

- Be conservative and, if needed, leave some of the upper phase behind to avoid carrying over chloroform contamination or white cloudy liquid from the interface between the aqueous and organic phases

Warnings:

- Don't disturb the interface!
- If the aqueous phase has a noticeable colour, you need to repeat the Ph:Ch:Is steps

DNA precipitation

1. Add 1/10 volume (~60ul) of 3M Sodium Acetate (pH8) for a concentration of 0.3M after mixing with the sample.

2. Mix well by inverting the tube.

3. Immediately after, add 1 volume (~600ul) of ice cold isopropanol.

4. Mix it by gently inverting the tube multiple times.

5. Incubate the samples for one and a half hours at 4C (fridge)

DNA cleaning and elution

1. Cool the centrifuge by selecting 5C and pre-centrifuging for 20 minutes without samples

2. Centrifuge the samples at 5C and 15,000g for 20 minutes.

3. Place a tube with dd molecular water (clean and autoclave) in the incubator at 37C, or in the heating block at 37C.

4. Carefully remove the tubes from the centrifuge. Identify the pellet (if visible, it will be on the same side of the tube as the hinge - centrifugal forces!).

5. Without disturbing the pellet, dispose of the isopropanol in a beaker.

6. Add 500ul of freezer cold 70% ethanol to each sample, invert the tube gently a few times.

7. Centrifuge at 12,000g for 15 minutes at 5C

Warnings:

- Place them in the centrifuge with the hinge of the tube pointing outwards.

8. Without disturbing the pellet, dispose of the ethanol in a beaker

> **Suggestions:**
> - Do it slowly, the pellet contains your DNA and you don’t want it to dislodge from the tube wal

9. Repeat wash one more time with 90% ethanol

10. Spin down the tubes to push to the bottom any residual ethanol remaining in the tube walls.

11. If you can see the pellet, you can use a 10ul tip to remove the remaining ethanol as long as you don’t touch the walls or the pellet.

> **Warnings:**
> - If you don’t see the pellet, don't try doing it.

12. Let the tubes with the lid open, ideally in the hood with laminar flow going

13. Incubate at room temperature for 15 minutes to allow for remaining ethanol to evaporate

> **Suggestions:**
> - You can monitor the pellet, if it looks dry and there are no droplets in the tube’s wall, you can proceed

> **Warnings:**
> - Don’t over dry the pellet as it will be hard to resuspend the DNA.

14. Add 35ul of the pre-heated molecular water (37C) to each tube to resuspend the pellet

15. Incubate at 37C for 15 minutes or more (not longer than half hour) until you no longer see the pellet (all DNA should be resuspended into the water)

16. Proceed to DNA quantification or long-term storage at -20C